Lab Report – EE-170L Computer Programming (Lab 13)

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DATE	12/5/2026

Objectives

- To understand the concept of structure**s** in C++.
- To learn how to declare and initialize structures.
- To practice accessing structure members using dot notation.
- To apply structures for storing and displaying real-world data.

Theory

- A **s**tructur**e** is a collection of variables (members) of different data types grouped under one name.
- **Syntax:**

```
struct StructName {
    int id;
    float value;
    char name[20];
};
```

- Unlike arrays (same type only), structures can hold mixed types.
- Structures can be nested, copied, and pa**ss**ed **to** functions.
- Members are accessed using the dot operator **(.)**.

Lab Tasks

Task 1: Structure Declaration and Initialization

```
#include <iostream>      // tell the compiler to include input output stream library

using namespace std;     // tells the compiler to use standard namespace


struct Person {            // structure declaration

   string name;           // member: name

   int age;               // member: age

   string address;        // member: address

};


int main() {               // main function starting point

   Person person1 = {"Naeem ul Haq", 19, "Peshawar"};      // initialize structure

   cout << "Name: " << person1.name << endl;                // display name

   cout << "Age: " << person1.age << endl;                  // display age

   cout << "Address: " << person1.address << endl;          // display address
```

```
    return 0;                    // tells the compiler that progrsm has ended
successfully

}
```

Task 2: Student Structure

```
#include <iostream>       // tell the compiler to include input output stream
library

using namespace std;      // tells the compiler to use standard namespace



struct Student {            // structure declaration

    string name;            // member: name

    int age;                // member: age

    char grade;             // member: grade

};


int main() {                                          // main function starting
point

    Student student1 = {"Ali Khan", 20, 'A'};          // initialize student

    cout << "Name: " << student1.name << endl;      // output message

    cout << "Age: " << student1.age << endl;          // output message
```

```
    cout << "Grade: " << student1.grade << endl;  // output message

    return 0;          // tells the compiler that program has ended successfully

}
```

Task 3: Accessing Structure Elements

```
#include <iostream>     // tell the compiler to include input output stream
library

using namespace std;  // tells the compiler to use standard namespace


struct Book {                              // structure declaration

    string title;                         // member: title

    string author;                         // member: author

    float price;                          // member: price

    int pages;                            // member: pages

};


int main() {                              // main dunction starting point

 Book book1 ={"C++ Programming", "Bjarne Stroustrup", 999.99, 1200};
//struc1

 Book book2 = {"Data Structures", "Mark Weiss", 750.50, 850};  //struc2
```

```
    cout << "Book 1: " << book1.title << ", " << book1.author     //output
message
        << ", Price: " << book1.price << ", Pages: " << book1.pages << endl;

    cout << "Book 2: " << book2.title << ", " << book2.author    //output message
        << ", Price: " << book2.price << ", Pages: " << book2.pages << endl;

    return 0;              // tells the compiler that program has ended
successfully
}
```

Results

- Successfully declared and initialized structures with multiple members.
- Demonstrated accessing structure members using dot notation.
- Showed how structures can store different types of data together.
- Applied structures to real-world examples (Person, Student, Book).